

Multiproposal Elliptical Slice Sampling

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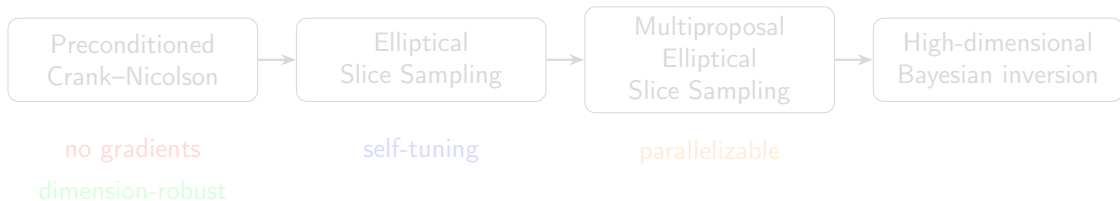
SIAM Uncertainty Quantification

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Summary

Goal: Scalable MCMC for high-dimensional posteriors with form

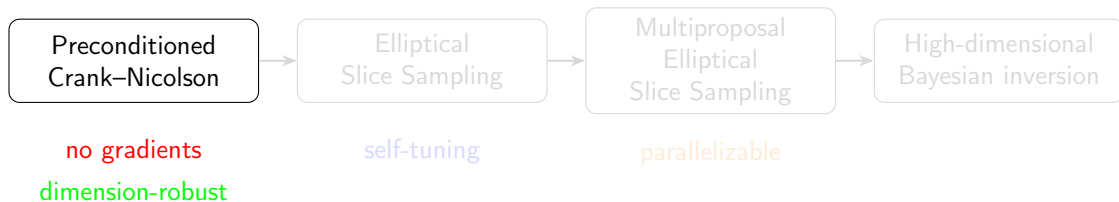
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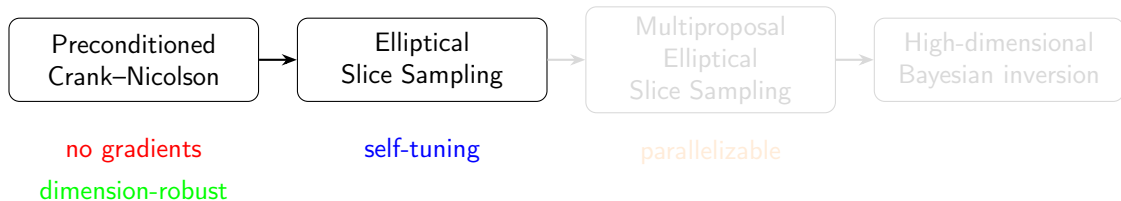
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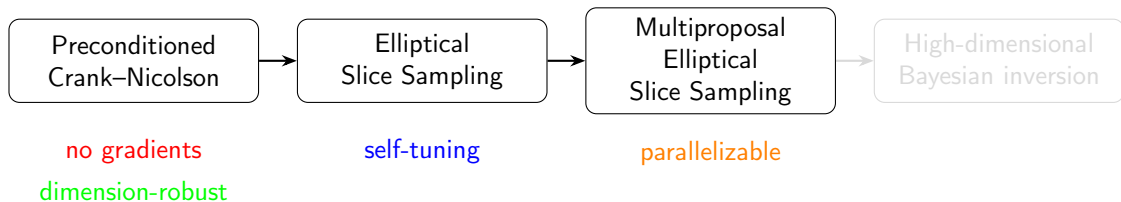
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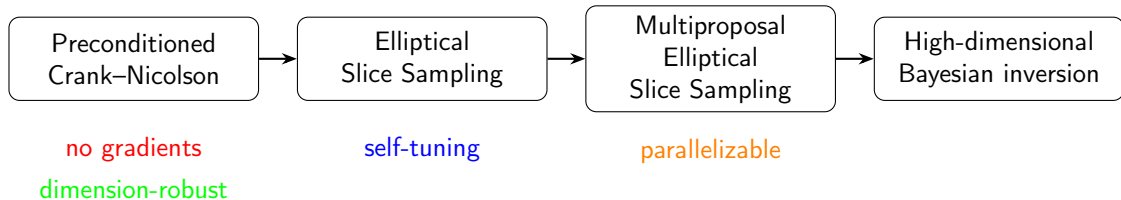
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Motivation

- Bayesian inverse problem

$$\pi(x|y) \propto \pi(y|x) \pi(x), \quad y \in \mathbb{R}^k \text{ (observations)}, \quad x \in \mathbb{R}^d \text{ (parameters)}$$

- Often non-parametric:

$$\{x(s) : s \in D\}, \quad D \subset \mathbb{R}^d.$$

- Sampling with Markov models:

- Construct ergodic Markov chain $X = \{X_1, X_2, \dots, X_N\}$ with π as invariant measure.
- Approximate expectations:

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{i=1}^N f(X_i) = \hat{\mathbb{E}}_{\pi}[f(X)]$$

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Markov chain Monte Carlo (MCMC)

Algorithm 1: Metropolis-Hastings (Hastings, 1970)

Input: Current state x_i , target density $\pi(x)$, proposal density $q(\cdot|x)$.

1. Sample $x^* \sim q(\cdot|x_i)$
2. Compute acceptance probability

$$\alpha(x_i, x^*) = \min\left\{1, \frac{\mathcal{L}(y|x^*)}{\mathcal{L}(y|x_i)} \times \frac{\pi(x^*)}{\pi(x_i)} \times \frac{q(x_i|x^*)}{q(x^*|x_i)}\right\}$$

3. Accept x^* with probability $\alpha(x_i, x^*)$, otherwise reject.
-

total cost _{ϵ} = cost per iteration $\times$ number of iterations _{ϵ} .

- Both increase with input size.

Pre-conditioned Crank-Nicolson (pCN)

Target¹:

$$\pi(x|y) \propto \mathcal{L}(y|x) \mathcal{N}(x; 0, C).$$

Algorithm 2: pCN (Cotter et al., 2013; Neal, 1999)

Input: Current state x , hyperpar $\rho \in (0, 1)$.

1. Propose $x^* = \rho x + \sqrt{1 - \rho^2} \nu$, $\nu \sim \mathcal{N}(0, C)$ no gradients
 2. Compute $\alpha(x, x^*) = \min\{1, \mathcal{L}(y|x^*)/\mathcal{L}(y|x)\}$
 3. Accept x^* with probability $\alpha(x, x^*)$, otherwise reject.
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Proposal is reversible with respect to the prior:

$$\mathcal{N}(x; 0, C) \mathcal{N}(x^*; \rho x, (1 - \rho^2)C) = \mathcal{N}(x^*; 0, C) \mathcal{N}(x; \rho x^*, (1 - \rho^2)C) \Rightarrow \text{dimension-free}$$

¹Stuart (2010); Dashti and Stuart (2013)

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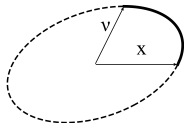
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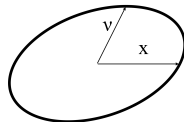
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- Elliptical proposal:

$$x^* = x \cos \theta + \nu \sin \theta, \quad \nu \sim \mathcal{N}(0, C), \quad \theta \in (0, 2\pi].$$



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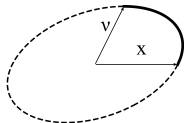
(b) ESS proposal

- Proposal holds the prior invariant.
- We can do better than elliptical MH $\rightarrow$ [self-tuning](#).

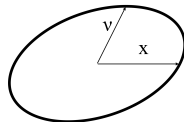
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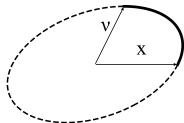
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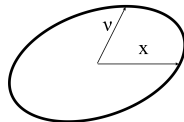
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Slice Sampling

- Augmented variable method.
- Idea:

$$\pi(x, y) = \begin{cases} \frac{1}{\int f(x) dx} & 0 < y < f(x) \\ 0 & \text{otherwise} \end{cases}$$

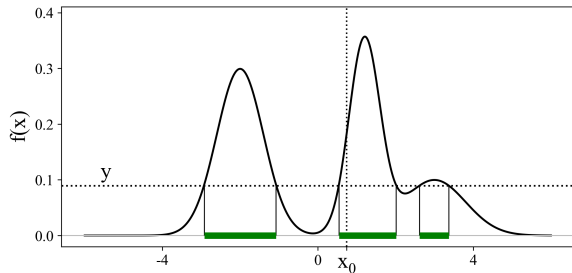


Figure: Slice sampling in 1D.

Algorithm 2: Slice sampling (Neal, 2003)

1. Sample $y \sim \mathcal{U}(0, f(x))$, defining *slice*
 $\mathcal{S} = \{x : f(x) \geq y\}$
 2. Sample $x \sim \mathcal{U}(\mathcal{S})$
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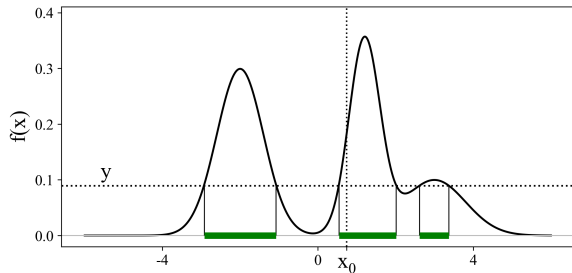


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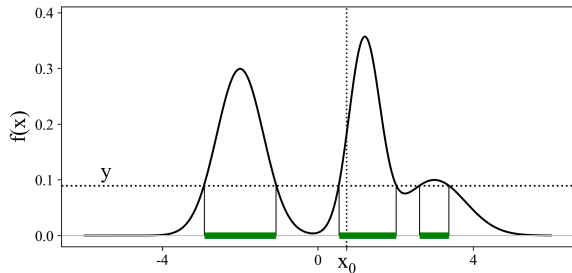


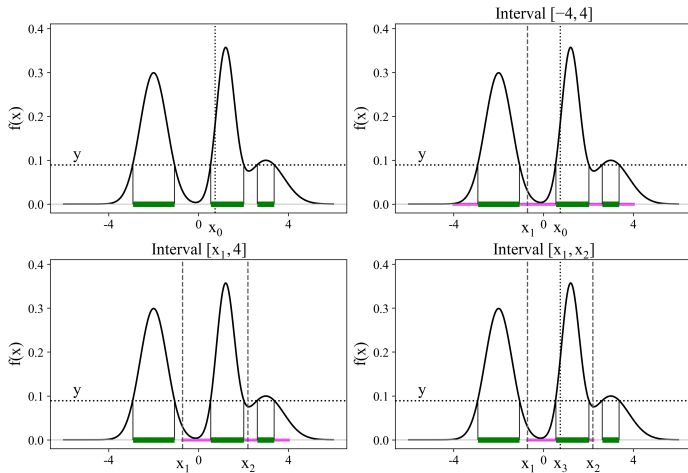
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Shrinking

- How to sample $x \sim \mathcal{U}(\mathcal{S})$ when $\mathcal{S}$ is unknown?



Elliptical Slice Sampling (continued)

Algorithm 4: ESS (single iteration)

Input: Current state x .

Draw $\nu \sim \mathcal{N}(0, C)$

Draw $\theta_x \sim \mathcal{U}(0, 2\pi)$

Draw slice level $y \sim \mathcal{U}(0, \mathcal{L}(y|x))$

Set $\theta_{\min} \leftarrow 0, \theta_{\max} \leftarrow 2\pi$

while *true* **do**

 Sample $\theta \sim \mathcal{U}(\theta_{\min}, \theta_{\max})$

 Propose $x^* = x \cos \theta + \nu \sin \theta$

if $\mathcal{L}(y|x^*) \geq y$ **then**

 accept $x \leftarrow x^*$ and stop

else

 shrink $(\theta_{\min}, \theta_{\max})$ around θ_x and continue [self-tuning](#)

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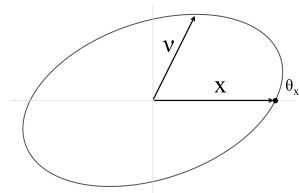
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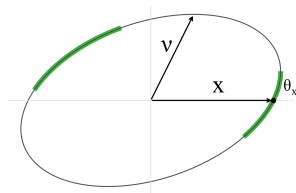
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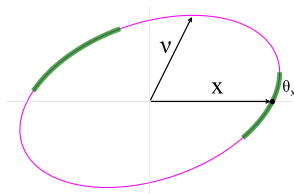
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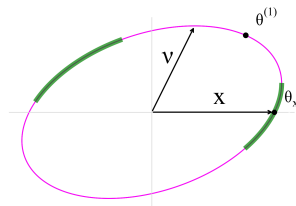
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One iteration of Elliptical Slice Sampling

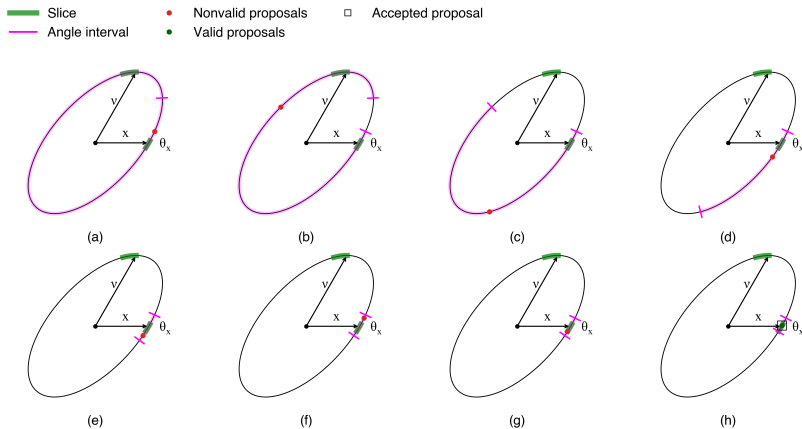


Figure: One iteration of ESS.

Mess in action

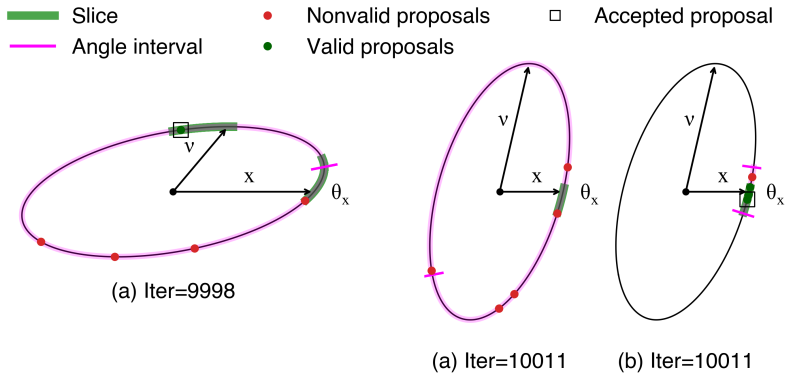


Figure: Two iterations of MESS with $M = 5$.

Multiproposal Elliptical Slice Sampling

Algorithm 5: MESS (single iteration)

Input: Current state x , number of proposals M .

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while true do

 Draw $\theta^{(j)} \sim \mathcal{U}(\theta_{\min}, \theta_{\max})$, $j = 1, \dots, M$ parallelizable

 Propose $x^{(j)} = x \cos \theta^{(j)} + \nu \sin \theta^{(j)}$

 Compute $\mathcal{A} = \{j : \mathcal{L}(y|x^{(j)}) \geq y\}$

if $|\mathcal{A}| = 0$ **then**

 shrink using all M angles and continue

else

 select $m \in \mathcal{A}$, set $x \leftarrow x^{(m)}$ and stop.

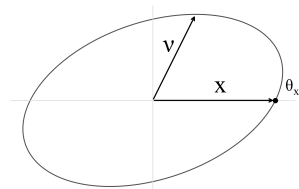


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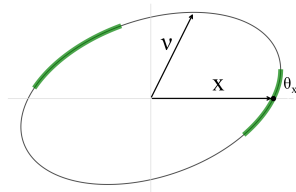


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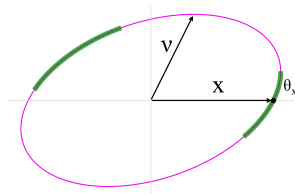


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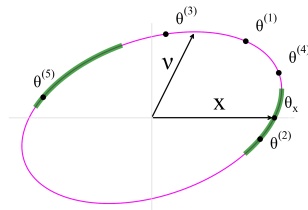


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Acceptance step with transition matrix

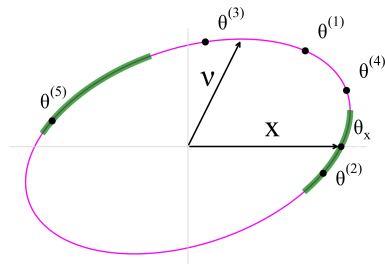
At subiteration k :

- Sample $\theta^{(1)}, \dots, \theta^{(5)}$.
- Valid indices: $\mathcal{A} = \{j : \mathcal{L}(\theta^{(j)}) \geq y\} = \{2, 5\}$.
- **Ordered** valid angles:

$(\theta^{(2)}, \theta_x, \theta^{(5)})$ with labels $(0, 1, 2)$.

- Transition matrix with entries

$$P = \begin{bmatrix} 0 & P_{01} & P_{02} \\ P_{10} & 0 & P_{12} \\ P_{20} & P_{21} & 0 \end{bmatrix}$$



Acceptance step with transition matrix

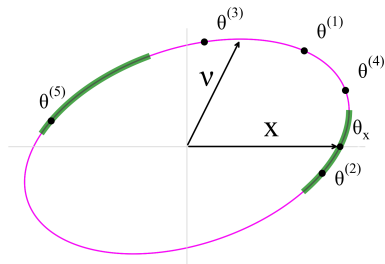
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- Transition matrix with entries

$$P = \begin{bmatrix} 0 & P_{01} & P_{02} \\ P_{10} & 0 & P_{12} \\ P_{20} & P_{21} & 0 \end{bmatrix}$$



Acceptance step with transition matrix

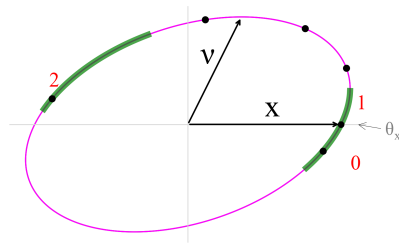
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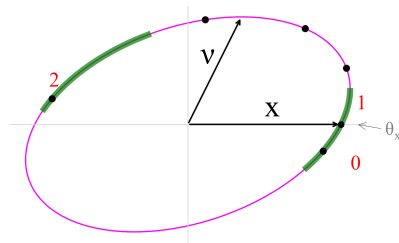
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Constructing the transition matrix

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$$\mathbf{P}^{\text{unif}} = \begin{bmatrix} 0 & 0.5 & 0.5 \\ 0.5 & 0 & 0.5 \\ 0.5 & 0.5 & 0 \end{bmatrix}$$

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$$\max_{\mathbf{P}} \sum_{r=1}^3 \sum_{s=1}^3 D_{\text{ang}}(r, s) P_{rs}$$

subject to double-stochastic $\mathbf{P}$ with zero diagonal.

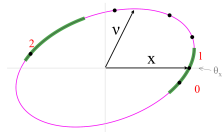


Figure: Labels of ordered valid angles.

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$$D_{\text{ang}} = \begin{bmatrix} 0 & 0.3 & 3 \\ 0.3 & 0 & 2.7 \\ 3 & 2.7 & 0 \end{bmatrix} \quad (\text{rad})$$

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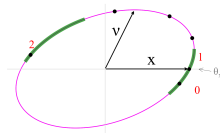


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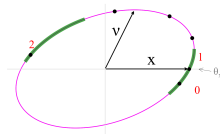


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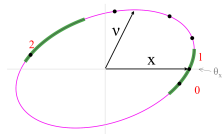


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Mixing improvement with uniform MESS

- Average number of shrinking steps decays with M :

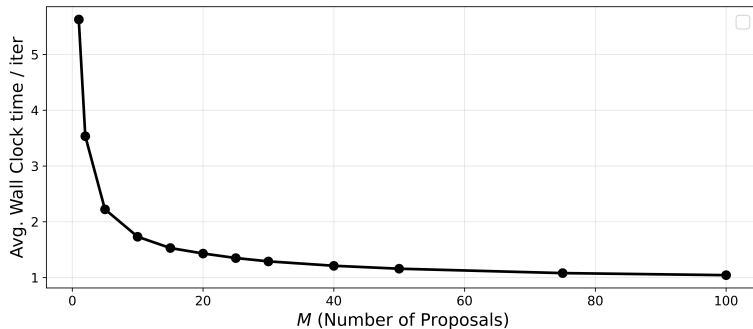


Figure: Logistic regression model from Murray et al. (2010).

- With M cores, wall-clock time decreases $\approx$ shrinking steps.

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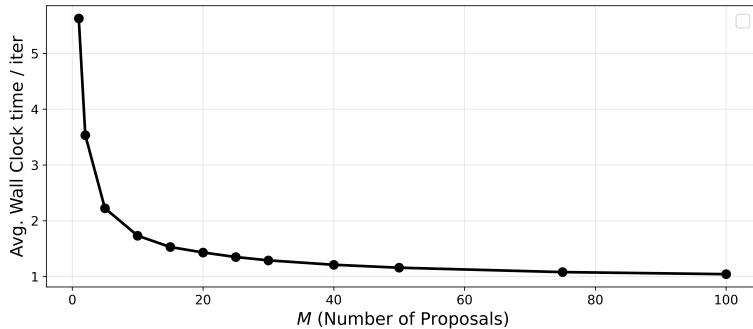


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$$\frac{d}{dt}\Theta + (A + \kappa I)\Theta = g.$$

- $\Theta \in \mathbb{R}^d$: Solute concentration (in Fourier basis).
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- Gaussian prior on the non-zero elements of A :

$$A \sim N(0, C), \quad C = \text{diag}(\{\text{Var}(a_{ij})\}).$$

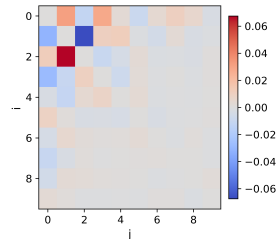
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- C penalizes long range interactions and high frequencies.

- Gaussian likelihood:

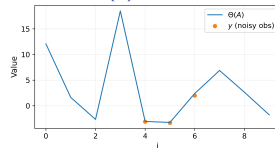
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(b) $\Theta(A_{10})$ and y_3 .

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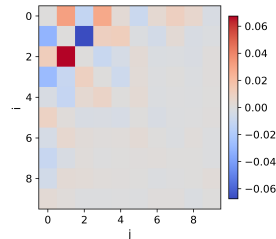
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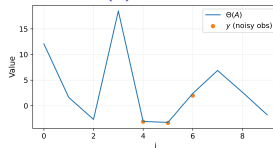
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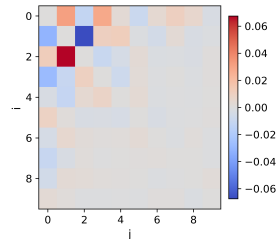
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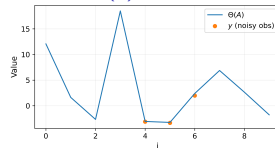
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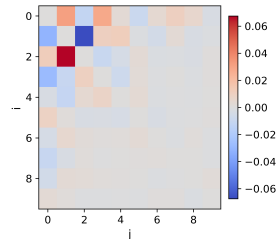
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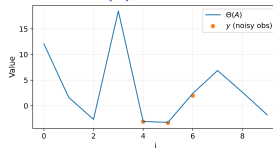
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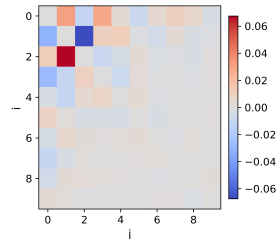
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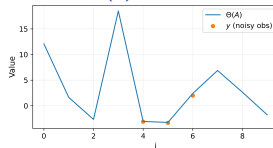
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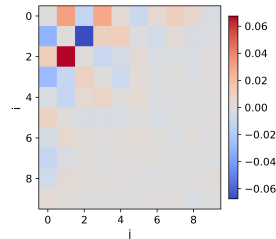
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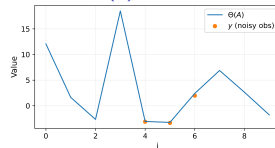
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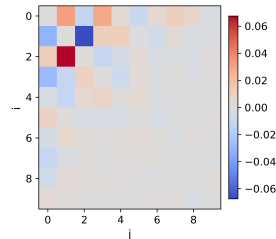
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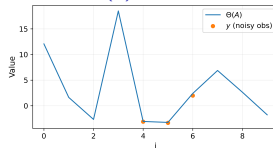
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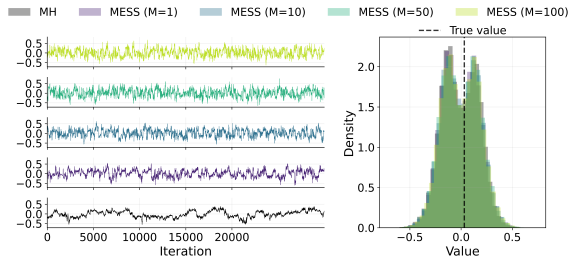


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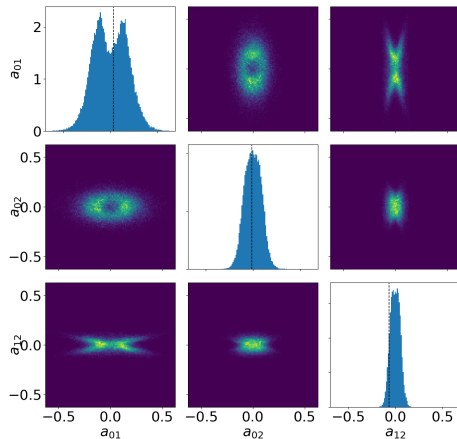


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Posterior distribution for $d = 10$

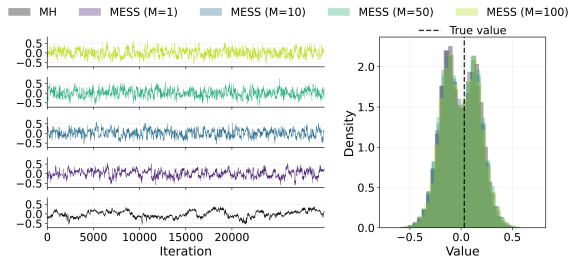


(a) Traceplots for a_{01} .

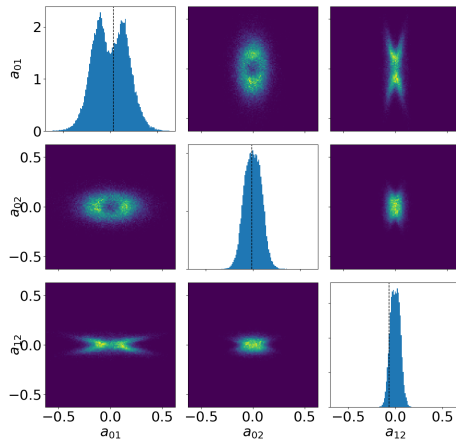


(b) Posterior marginals and pairwise density plots at $d = 10$.

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Scaling with dimension

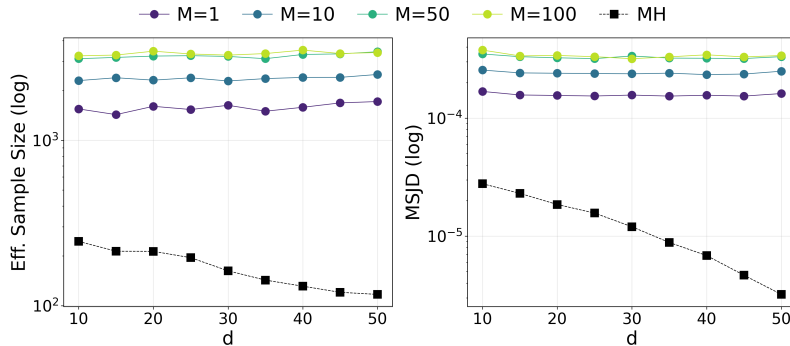


Figure: MESS mixing rate does not degrade with dimension.

Conclusions

Multiproposal Elliptical Slice Sampling (MESS) is

Self-tuning

no gradients

dimension-robust

parallelizable.

Useful for large-scale and/or non-parametric Bayesian estimation with Gaussian priors.



References I

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Comparison with Gibbs and HMC: blind deconvolution inverse problem

- Observation model:

$$d = w \star c + e, \quad w \sim N(0, C_w), \quad c \sim N(0, C_c), \quad e \sim N(0, C_e).$$

- Posterior is multimodal (sign-shift symmetry).

